VIEW OF DATA

- *DATA MODELS
- *DATA ABSTRACTION
- *INSTANCES AND SCHEMA

View of Data

- A database system is a collection **of interrelated data and a set of programs that allow users to access and modify these data**.

- A major purpose of a database system is to provide users with an **abstract view** of the data.

View

- In a Database Management System (DBMS), a view is a **virtual table based on the result of an SQL query.**

- It **doesn't store any data itself** but presents a customized subset of data from one or more base tables.

- Views are used to simplify data access, enhance security, and provide a tailored view of the database for different users or applications.

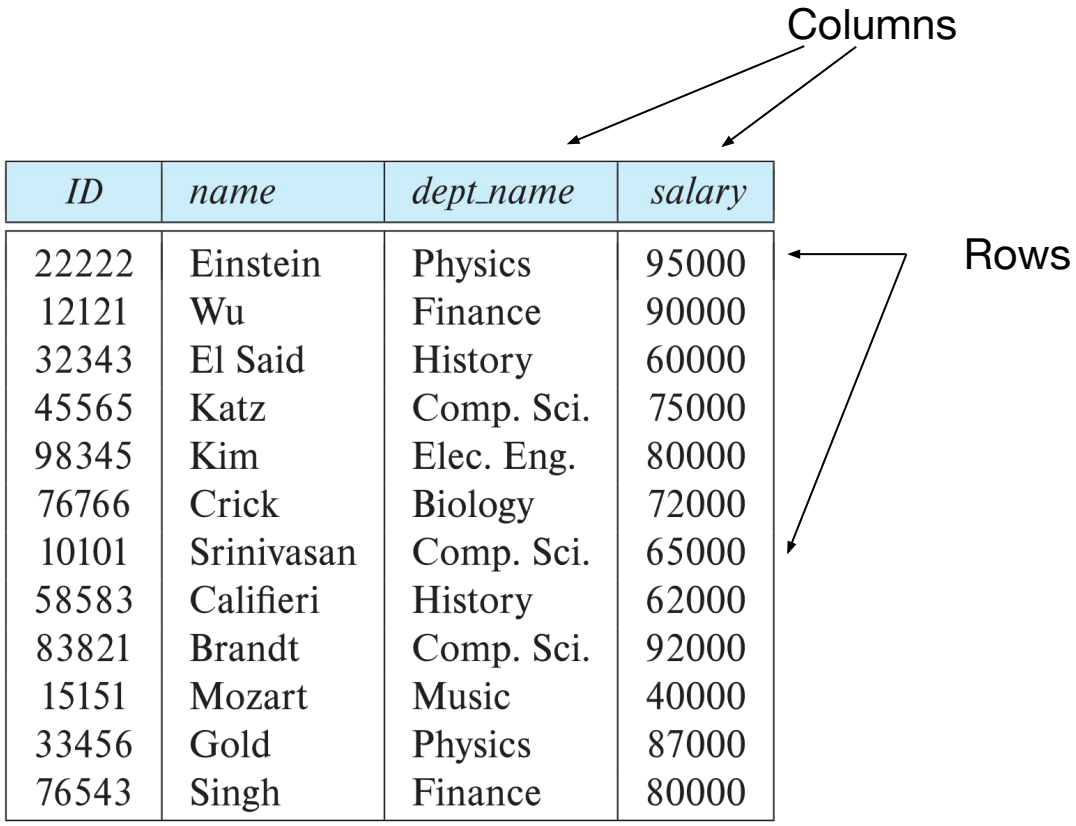
Data Models

- A **blueprint** that defines how data is structured, organized, and accessed within a database.
- It provides a way to represent **real-world entities and their relationships**, ensuring **data consistency and efficient retrieval.**
- A collection of conceptual tools for describing
    - Data
    - Data relationships
    - Data semantics
    - Data constraints

- Types of Data Models
    - Relational model
    - Entity-Relationship data model (mainly for database design)
    - Object-based data models (Object-oriented and Object-relational)
    - Semi-structured data model (XML)
    - Other older models:
        - Network model
        - Hierarchical model

Data Models - Relational Model

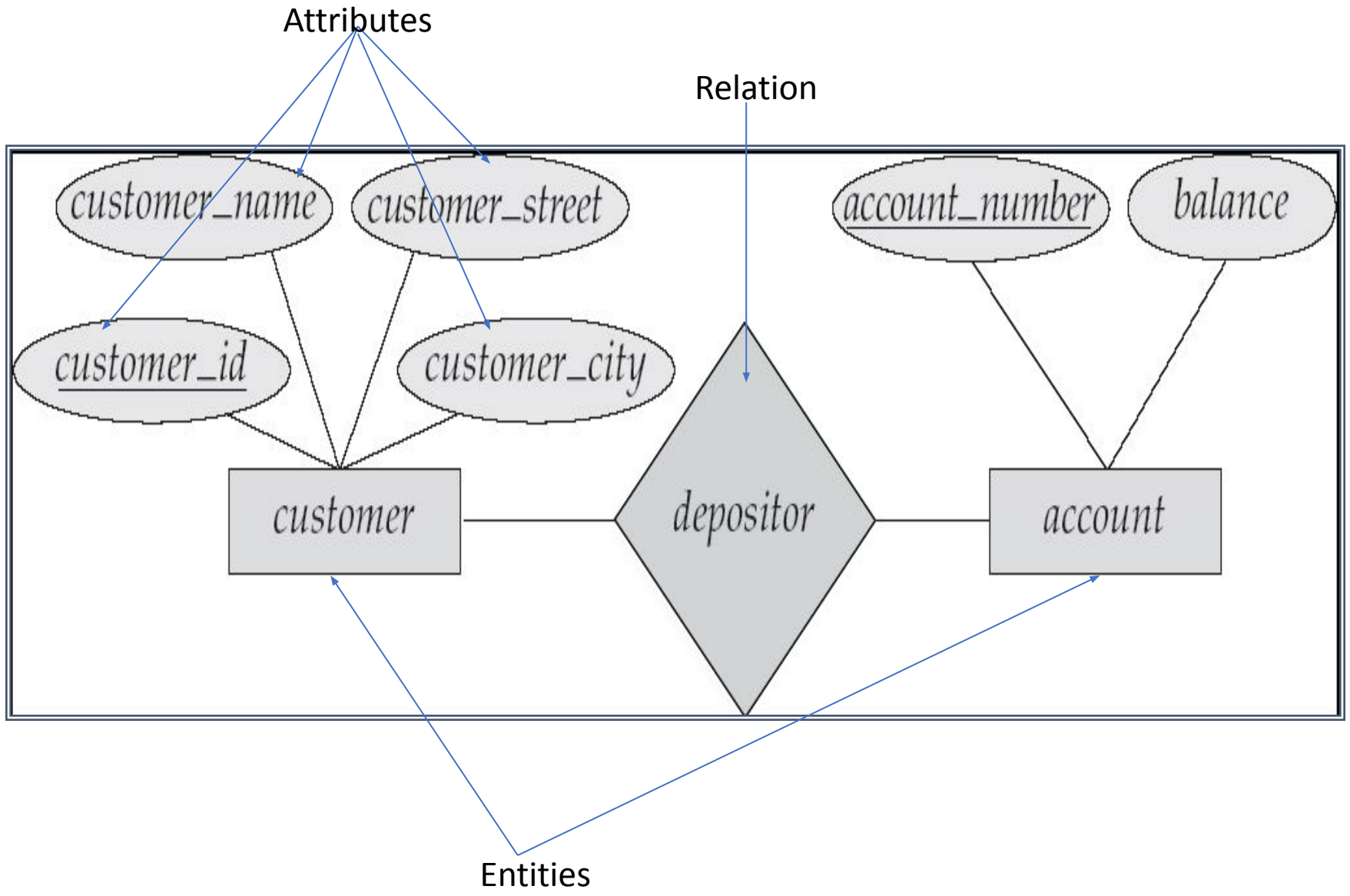
- All the data is stored in various tables.
- Example of tabular data in the relational model



ID	*name*	*dept_name*	*salary*
22222	Einstein	Physics	95000
12121	Wu	Finance	90000
32343	El Said	History	60000
45565	Katz	Comp. Sci.	75000
98345	Kim	Elec. Eng.	80000
76766	Crick	Biology	72000
10101	Srinivasan	Comp. Sci.	65000
58583	Califieri	History	62000
83821	Brandt	Comp. Sci.	92000
15151	Mozart	Music	40000
33456	Gold	Physics	87000
76543	Singh	Finance	80000

(a) The *instructor* table

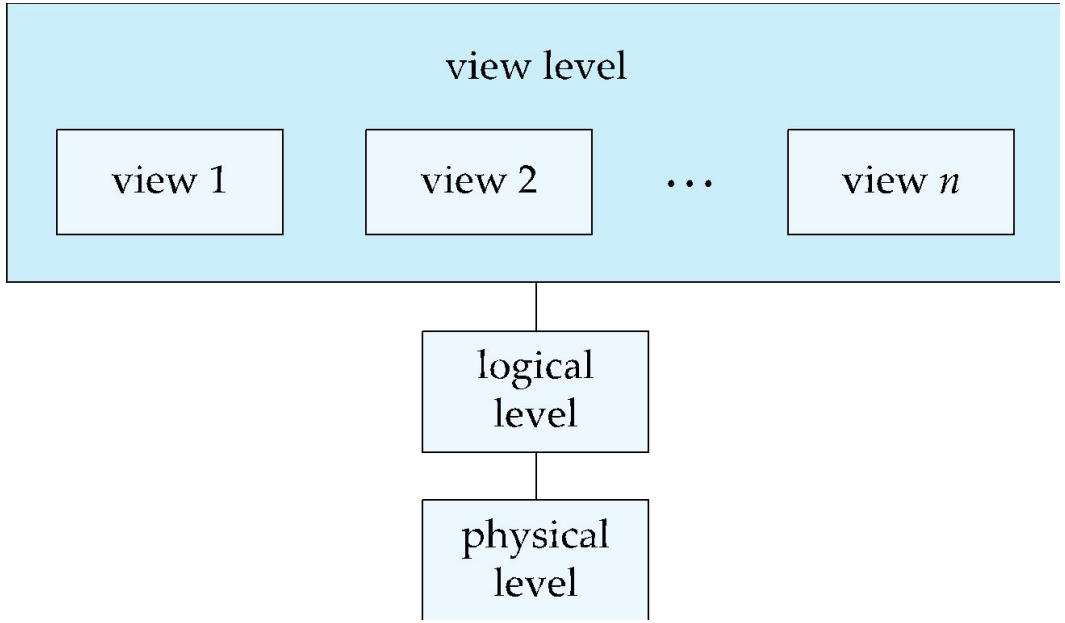
Data Models - ER Model



Levels of Data Abstraction

Data abstraction

- The process of hiding complex details of how data is stored and managed
- Presenting users with a simplified view of the data
- Allows users to interact with the database without the need to understand the underlying implementation details.

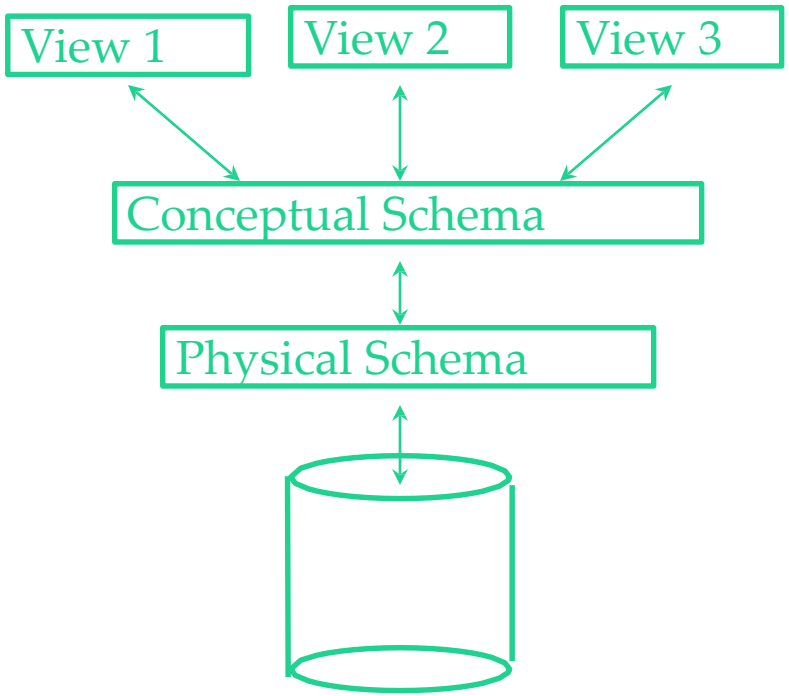


Levels of Data Abstraction

- Many *views*
  - Views describe how users see the data.

- Single *conceptual (logical) schema*
  - Conceptual schema defines
  - logical structure

- Single *physical schema*:
  - Physical schema describes the files and indexes used.



Levels of Data Abstraction

- **Physical level**:
  - describes how the data is physically stored in the database (e.g., file structures, storage devices)
  - It's the lowest level of abstraction and is hidden from most users.

- **Logical level**:
  - This level defines the structure of the database, including tables, attributes, relationships, and constraints.
  - It's the level at which users interact with the database through queries and other operations.

- **View level**:
  - This level provides customized views of the data for specific users or applications.
  - It allows users to see only the data relevant to them, enhancing security and simplifying data access.

- **Example:**
  - Imagine using a search engine to find information. You don't need to know how the search engine stores and retrieves data (physical level), or even the exact structure of the database (logical level). You only need to know how to enter your search query (view level).

Schemas and Instances

- In a Database Management System (DBMS), schemas and instances are two fundamental concepts that describe the **structure and content** of a database

Schema

- A schema is the overall description, design, or blueprint of a database
- It defines the logical structure of a database, including:
  - Tables.
  - Columns/fields and their data types.
  - Relationships between tables (e.g., primary and foreign keys).
  - Constraints that govern the data (e.g., unique, not null)

- **Example**
  - It's like the architectural plan of a building, specifying the number of rooms, their layout, and how they connect, but not containing the actual furniture or people living in them.

Schemas and Instances

- Schemas are designed by **Database Administrators (DBAs)** using **Data Definition Language (DDL) commands** like CREATE TABLE, ALTER TABLE, and DROP TABLE.

- There are three types of schema based on the level of abstraction:
  - **Physical Schema (Internal Schema):**
    - Describes how data is physically stored in storage devices, including details about files, indexes, and storage structures.
  - **Logical Schema (Conceptual Schema):**
    - Defines the logical structure of the database, including tables, relationships, and constraints, independent of physical storage details.
  - **External Schema (View Schema):**
    - Defines the database views relevant to specific users or applications, representing how data is visible to them.

Schemas and Instances

Instances

- An instance is the collection of actual data stored in the database at a particular point in time
- Instances are dynamic and change frequently as data is inserted, updated, or deleted using **Data Manipulation Language (DML) commands** like INSERT, UPDATE, and DELETE
- An instance must conform to the structure defined by the schema
- **Example**
  - It's like the current state of a building, with furniture in the rooms and people occupying them, representing a snapshot of the data at that moment

- **Relationship between schema and instance**
  - The schema is the blueprint, while the instance is the actual data that adheres to that blueprint.
  - The schema remains relatively stable over time, changing only when the database's structure or design is modified.
  - The instance, however, changes frequently as data is added, deleted, or updated in the database.

Physical Data Independence

- Physical data independence in a Database Management System (DBMS) refers to the ability to **modify the physical storage structure of the database without requiring changes to the logical schema (conceptual level) or the external schemas (view level)**.

- This means that how the data is physically stored on the hardware can be altered without affecting the way users or applications interact with the database

- Physical storage modification includes
  - **changing storage devices**
  - **modifying indexing strategies**
  - **optimizing data storage through compression or partitioning**
  - **altering the database's file organization**
  - **changing the location of the database files**

DATABASE LANGUAGES / COMPONENTS OF SQL

*DATA DEFINITION LANGUAGE(DDL)

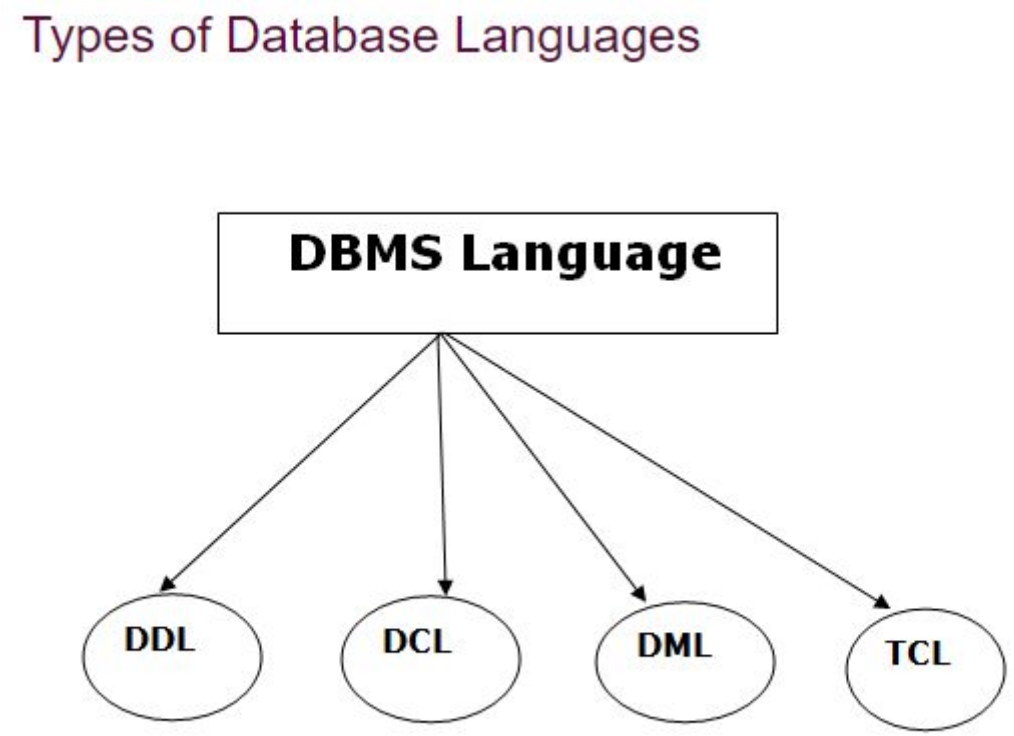
*DATA MANIPULATION LANGUAGE (DML)

*DATA CONTROL LANGUAGE (DCL)

* TRANSACTION CONTROL LANGUAGE(TCL)

Database Languages in DBMS

- A DBMS has appropriate languages and interfaces to express database queries and updates.
- Database languages can be used to read, store and update the data in the database.



1. Data Definition Language (DDL)

- **DDL** stands for **D**ata **D**efinition **L**anguage. It is used to define database structure or pattern.
- It is used to create schema, tables, indexes, constraints, etc. in the database.
- Using the DDL statements, you can create the skeleton of the database.
- Data definition language is used to store the information of metadata like the number of tables and schemas, their names, indexes, columns in each table, constraints, etc.

Here are some tasks that come under DDL:

- **Create:** It is used to create objects in the database.
- **Alter:** It is used to alter the structure of the database.
- **Drop:** It is used to delete objects from the database.
- **Truncate:** It is used to remove all records from a table.
- **Rename:** It is used to rename an object.
- **Comment:** It is used to comment on the data dictionary.

These commands are used to update the database schema that's why they come under Data definition language.

Data Definition Language (DDL)

- Specification notation for defining the database schema

  Example: **create table** *instructor* (
  ID **char**(5),
  name **varchar**(20),
  dept_name **varchar**(20),
  salary **numeric**(8,2))

- DDL compiler generates a set of table templates stored in a ***data dictionary***
- Data dictionary contains metadata (i.e., data about data)
  - Database schema
  - Integrity constraints
    - Primary key (ID uniquely identifies instructors)
  - Authorization
    - Who can access what

2. Data Manipulation Language (DML)

DML stands for **D**ata **M**anipulation **L**anguage. It is used for accessing and manipulating data in a database. It handles user requests.

- DML also known as query language

- There are basically two types of data-manipulation language
  - **Procedural DML** -- require a user to specify what data are needed and how to get those data.
  - **Declarative DML** -- require a user to specify what data are needed without specifying how to get those data.
- Declarative DMLs are usually easier to learn and use than are procedural DMLs.
- Declarative DMLs are also referred to as non-procedural DMLs
- The portion of a DML that involves information retrieval is called a **query language.**

Procedural:

1. Go to kitchen
2. Get sugar, milk, and tea,
3. Mix them, and heat over the fire till it boils
4. Put that in a cup and bring it to me

Declarative:

1. Get me a cup of tea.

2. Data Manipulation Language (DML)

- **Select:** It is used to retrieve data from a database.
- **Insert:** It is used to insert data into a table.
- **Update:** It is used to update existing data within a table.
- **Delete:** It is used to delete all records from a table.
- **Merge:** It performs UPSERT operation, i.e., insert or update operations.
- **Call:** It is used to call a structured query language or a Java subprogram.
- **Explain Plan:** It has the parameter of explaining data.
- **Lock Table:** It controls concurrency.

3. Data Control Language (DCL)

- **DCL** stands for **D**ata **C**ontrol **L**anguage. It is used to retrieve the stored or saved data.
- The DCL execution is transactional. It also has rollback parameters.

  (But in Oracle database, the execution of data control language does not have the feature of rolling back.)

Here are some tasks that come under DCL:

- **Grant:** It is used to give user access privileges to a database.
- **Revoke:** It is used to take back permissions from the user.

There are the following operations which have the authorization of Revoke:

CONNECT, INSERT, USAGE, EXECUTE, DELETE, UPDATE and SELECT.

4. Transaction Control Language (TCL)

TCL is used to run the changes made by the DML statement. TCL can be grouped into a logical transaction.

Here are some tasks that come under TCL:

- **Commit:** It is used to save the transaction on the database.
- **Rollback:** It is used to restore the database to original since the last Commit.

SQL Query Language

- SQL is a standard language for storing, manipulating and retrieving data in databases.
- SQL query language is nonprocedural. A query takes as input several tables (possibly only one) and always returns a single table.
- Example to find all instructors in Comp. Sci. dept

  select *name*
  from *instructor*
  where *dept_name* = 'Comp. Sci.'

- SQL is **NOT** a Turing machine equivalent language
- To be able to compute complex functions SQL is usually **embedded in some higher-level language**
- Application programs generally access databases through one of
  - Language extensions to allow embedded SQL
  - Application program interface (e.g., ODBC/JDBC) which allow SQL queries to be sent to a database